

INDIAN INSTITUTE OF TECHNOLOGY ROPAR

Introduction to Stochastic Calculus

Lecture Notes

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These notes cover: Ordinary Calculus Review · Riemann-Stieltjes Integration ·
Stochastic Integrals · Itô's Formula · Geometric Brownian Motion · OU Process

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1 Introduction to Stochastic Calculus

Classical calculus deals with deterministic functions that vary smoothly over time. In many real-world applications — particularly in finance, physics, and biology — we encounter quantities that evolve *randomly*. Stochastic calculus provides the mathematical machinery to handle such random processes rigorously.

The central challenge is defining integrals of the form

$$\int_0^t f(s) dB(s) \quad \text{or} \quad \int_0^t X(s) dB(s),$$

where $B(t)$ is a **Brownian motion** (Wiener process).

Remark

The Brownian motion $B(t)$ has sample paths that are *continuous but nowhere differentiable*. This means there is no function $B'(t) = \frac{dB(t)}{dt}$ in the classical sense, and consequently the integral $\int_0^t g(s) dB(s)$ **cannot** be written as $\int_0^t g(s) B'(s) ds$. This is what makes stochastic calculus fundamentally different from ordinary calculus.

To develop a proper theory, we must first revisit the relevant concepts from ordinary calculus before building towards the stochastic integral.

2 Ordinary Calculus

2.1 Derivatives and Differentials

In classical analysis, the derivative of a function $G(t)$ is defined as

$$\frac{dG(t)}{dt} = \lim_{h \rightarrow 0} \frac{G(t+h) - G(t)}{h},$$

whenever this limit exists. When G is differentiable with derivative $g(t) = G'(t)$, the following three statements are completely equivalent:

Equivalence in Ordinary Calculus

Let $G(t)$ be a differentiable function. Then the following three statements are equivalent:

1. $\frac{dG(t)}{dt} = g(t)$ (derivative form)
2. $dG(t) = g(t) dt$ (differential form)
3. $G(t) = G(0) + \int_0^t g(s) ds$ (integral form)

An immediate consequence is that integrals with respect to a differentiable function G reduce to ordinary integrals:

$$\int_0^t f(s) dG(s) = \int_0^t f(s) g(s) ds.$$

2.2 The Riemann Integral

The Riemann integral is the standard tool for integrating well-behaved functions. Consider the **indicator (simple) function**:

$$f(t) = c \mathbf{1}_{(a,b]}(t) = \begin{cases} c & \text{if } t \in (a, b] \\ 0 & \text{otherwise.} \end{cases}$$

Its Riemann integral is given by:

$$\int_0^t f(s) ds = \int_0^t c \mathbf{1}_{(a,b]}(s) ds = \begin{cases} 0 & \text{if } t \leq a \\ c(t - a) & \text{if } a < t \leq b \\ c(b - a) & \text{if } t > b. \end{cases}$$

Geometrically, this is simply the area of the rectangle with height c and width $\min(t, b) - a$ (when $t > a$).

2.3 The Riemann–Stieltjes Integral

The Riemann–Stieltjes integral is a **generalization of the Riemann integral**, where we integrate with respect to an arbitrary function g (not just the identity $g(t) = t$).

Definition – Riemann–Stieltjes Integral

Let f and g be functions on $[0, T]$. Consider a partition

$$0 = t_0 < t_1 < \cdots < t_n = T$$

with intermediate points t_i^* satisfying $t_{i-1} \leq t_i^* \leq t_i$, and $\max_i |t_i - t_{i-1}| \rightarrow 0$ as $n \rightarrow \infty$. The **Riemann–Stieltjes integral** of f with respect to g is defined as:

$$\int_0^T f(t) dg(t) = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_i^*) [g(t_i) - g(t_{i-1})],$$

provided this limit exists.

2.3.1 Properties of the Riemann–Stieltjes Integral**Proposition – Existence Conditions**

- When $g(t) = t$, the Riemann–Stieltjes integral reduces to the ordinary **Riemann integral**.
- The integral exists if **either** f **or** g (or both) is **differentiable**.
- The integral exists if **either** f **or** g (or both) is of **bounded variation**.

Remark

Brownian motion $B(t)$ is *not* of bounded variation, and it is not differentiable. Therefore, the Riemann–Stieltjes integral $\int_0^t f(s) dB(s)$ is **not well-defined** in the classical sense. This motivates the need for Itô's stochastic integral.

2.3.2 Riemann–Stieltjes Integral for Simple Functions

For the indicator function $f(t) = c \mathbf{1}_{(a,b]}(t)$, the Riemann–Stieltjes integral with respect to a function g is:

$$\int_0^t f(s) dg(s) = \begin{cases} 0 & \text{if } t \leq a \\ c(g(t) - g(a)) & \text{if } a < t \leq b \\ c(g(b) - g(a)) & \text{if } t > b. \end{cases}$$

This is the natural analog of the Riemann case, simply replacing the length $t - a$ by the increment $g(t) - g(a)$.

3 Variation of Functions and Processes

Before introducing the stochastic integral, we need to understand two key concepts that explain *why* ordinary calculus breaks down for Brownian motion and how Itô's formula acquires its extra correction term: **bounded variation** and **quadratic variation**.

3.1 Bounded Variation

Background Knowledge: Bounded Variation

Definition. Let $f : [0, T] \rightarrow \mathbb{R}$. The **total variation** of f on $[0, T]$ is

$$V(f; [0, T]) = \sup_{\text{partitions}} \sum_{i=1}^n |f(t_i) - f(t_{i-1})|.$$

f is of **bounded variation** (BV) if $V(f; [0, T]) < \infty$.

Intuition. BV measures total up-and-down oscillation. A BV function cannot oscillate wildly — it is the difference of two monotone functions.

Examples:

- Every **monotone** function is BV, with $V = |f(T) - f(0)|$.
- Every **continuously differentiable** function is BV: $V(f; [0, T]) \leq \int_0^T |f'(t)| dt < \infty$.
- $f(t) = t \sin(1/t)$, $f(0) = 0$ is **not** BV on $[0, 1]$ — oscillations accumulate near 0.

Why it matters. The Riemann–Stieltjes integral exists when the integrator is BV. Brownian motion has **infinite total variation** almost surely: $V(B; [0, T]) = +\infty$. This is why a new integration theory (Itô's integral) is needed.

3.2 Quadratic Variation

Background Knowledge: Quadratic Variation

Definition. The **quadratic variation** of f on $[0, T]$ is the limit of squared increments:

$$[f, f]_T = \lim_{\|\Pi\| \rightarrow 0} \sum_{i=1}^n (f(t_i) - f(t_{i-1}))^2.$$

Smooth functions: zero quadratic variation. If f is C^1 , then

$$[f]_T \leq \max_i |f(t_i) - f(t_{i-1})| \cdot V(f; [0, T]) \rightarrow 0,$$

since the max increment $\rightarrow 0$ and total variation is finite. So $(df)^2 = 0$ — Taylor terms beyond first order vanish, giving the ordinary chain rule.

Brownian motion: non-zero quadratic variation. For $B(t)$: $[B, B]_T = T$ almost surely.

Derivation. With n equal steps of size T/n , let $Q_n = \sum_{i=1}^n (B(t_i) - B(t_{i-1}))^2$. Since each increment $\sim \mathcal{N}(0, T/n)$:

$$\mathbb{E}[Q_n] = n \cdot \frac{T}{n} = T, \quad \text{Var}(Q_n) = n \cdot 2 \left(\frac{T}{n} \right)^2 = \frac{2T^2}{n} \rightarrow 0.$$

So $Q_n \rightarrow T$ in L^2 (and a.s.). In differential notation: $\boxed{dB(t) \cdot dB(t) = dt}$.

3.3 The Itô Multiplication Table

The result $dB \cdot dB = dt$ leads to the **Itô multiplication table**:

Background Knowledge: The Itô Multiplication Table

$\times$	dt	$dB(t)$
dt	0	0
$dB(t)$	0	dt

Justification: $dt \cdot dt = 0$ (second-order infinitesimal); $dt \cdot dB \approx (dt)^{3/2} \rightarrow 0$ since $dB \approx \sqrt{dt}$; $dB \cdot dB = dt$ from the quadratic variation of Brownian motion.

Quadratic variation of a general Itô process. For $dX = Q dt + P dB$:

$$dX \cdot dX = (Q dt + P dB)^2 = Q^2(dt)^2 + 2QP dt dB + P^2(dB)^2 = P(t)^2 dt.$$

This identity $(dX)^2 = P(t)^2 dt$ is used directly in Itô's formula.

Remark

Summary of the distinction:

- A function of **bounded variation** (BV) has finite total oscillation and

zero quadratic variation. Ordinary Riemann–Stieltjes theory works for BV integrators.

- Brownian motion has **infinite total variation** but **finite quadratic variation** equal to t . This non-zero quadratic variation is precisely why Itô's formula has the extra $\frac{1}{2}P^2 f''$ correction term — the term that vanishes in ordinary calculus.

4 The Stochastic Integral

4.1 The Basic Stochastic Process

We now begin building the Itô stochastic integral. The construction starts with the simplest class of stochastic processes.

Definition – Basic Stochastic Process

The process X is called a **basic stochastic process** if it admits the representation

$$X_t(\omega) = C(\omega) \mathbf{1}_{(a,b]}(t), \quad a < b \in \mathbb{R},$$

where C is an $\mathcal{F}_a$ -measurable random variable.

The measurability condition $C \in \mathcal{F}_a$ means that C is determined by the information available up to time a . This is the **non-anticipation** (adaptedness) property that is crucial for the stochastic integral to be well-defined.

4.2 Stochastic Integral of a Basic Process

Definition – Itô Integral for Basic Processes

The stochastic integral of the basic process X with respect to Brownian motion B is defined as:

$$\left(\int_0^t X(s) dB(s) \right) (\omega) = \begin{cases} 0 & \text{if } 0 \leq t \leq a \\ C(\omega)(B(t, \omega) - B(a, \omega)) & \text{if } a < t \leq b \\ C(\omega)(B(b, \omega) - B(a, \omega)) & \text{if } t > b. \end{cases}$$

This definition mirrors the Riemann–Stieltjes construction for simple functions: we multiply the integrand value (frozen at the *left endpoint* a) by the increment of B over the interval. The use of the left endpoint is the hallmark of the Itô convention.

4.3 Useful Properties of the Itô Integral

Key Properties of the Itô Integral

Let $X(s)$ and $Y(s)$ be square-integrable adapted processes. Then:

(i) **Zero mean:**

$$\mathbb{E} \left[\int_0^t X(s) dB(s) \right] = 0.$$

(ii) **Itô Isometry:**

$$\mathbb{E} \left[\left(\int_0^t X(s) dB(s) \right)^2 \right] = \int_0^t \mathbb{E}[X(s)^2] ds.$$

(iii) **Cross-term isometry:**

$$\mathbb{E} \left[\int_0^t X(s) dB(s) \cdot \int_0^t Y(s) dB(s) \right] = \int_0^t \mathbb{E}[X(s)Y(s)] ds.$$

The **zero mean property** follows from the fact that Brownian increments have mean zero and are independent of $\mathcal{F}_a$ -measurable random variables. The **Itô isometry** is an analogue of Parseval's theorem and is fundamental in the L^2 theory of stochastic integration.

5 Itô's Formula

Itô's formula is the stochastic analogue of the chain rule in ordinary calculus. It is arguably the most important result in stochastic calculus.

5.1 The Chain Rule in Ordinary Calculus

For deterministic functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $x : [0, \infty) \rightarrow \mathbb{R}$, both continuously differentiable, the chain rule gives:

$$\frac{d}{dt} f(x(t)) = f'(x(t)) \frac{dx(t)}{dt},$$

or in differential form:

$$df(x(t)) = f'(x(t)) dx(t).$$

Using the Taylor series expansion:

$$df(x(t)) = f'(x(t)) dx(t) + \frac{1}{2!} f''(x(t)) (dx(t))^2 + \dots$$

In the deterministic setting, $(dx(t))^2$ corresponds to the **quadratic variation** of $x(\cdot)$. For smooth functions, higher-order terms vanish.

5.2 The Itô Process

Definition – Itô Process

An **Itô process** is a stochastic process $X(t)$ of the form

$$dX(t) = Q(t) dt + P(t) dB(t),$$

where $Q(t)$ is the **drift** coefficient and $P(t)$ is the **diffusion** coefficient (both adapted processes).

The critical difference from the deterministic case is the quadratic variation: for an Itô process,

$$dX(t) \cdot dX(t) = P(t)^2 dt.$$

This result follows directly from the Itô multiplication table (see Section 3 for the full derivation). Recall that for Brownian motion $dB \cdot dB = dt$ and $dt \cdot dt = 0$, so:

$\times$	dt	$dB(t)$
dt	0	0
$dB(t)$	0	dt

5.3 Itô's Formula (Time-Independent)

Theorem – Itô's Formula (1D)

Let $dX(t) = Q(t) dt + P(t) dB(t)$ be an Itô process, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable. Then:

$$df(X(t)) = \left[Q(t)f'(X(t)) + \frac{1}{2}P(t)^2 f''(X(t)) \right] dt + P(t)f'(X(t)) dB(t).$$

Derivation sketch: By the Taylor expansion,

$$df(X(t)) = f'(X(t)) dX(t) + \frac{1}{2}f''(X(t)) (dX(t))^2 + \text{higher order terms}.$$

Since variations of order ≥ 3 vanish and $(dX)^2 = P^2 dt$, substituting $dX = Q dt +$

$P dB$ gives:

$$\begin{aligned} df(X(t)) &= f'(X(t))(Q dt + P dB) + \frac{1}{2}f''(X(t)) \cdot P^2 dt \\ &= \underbrace{(Qf'(X) + \frac{1}{2}P^2f''(X))}_{\text{drift part}} dt + \underbrace{Pf'(X)}_{\text{diffusion part}} dB. \end{aligned}$$

Remark

The extra term $\frac{1}{2}P(t)^2f''(X(t)) dt$ (the **Itô correction term**) is unique to stochastic calculus. It arises because Brownian motion has non-zero quadratic variation. In ordinary calculus, $(dx)^2 = 0$ and this term disappears.

5.4 Itô's Formula (Time-Dependent)

Proposition – Extended Itô's Formula

For an Itô process $dX(t) = P(t) dB(t) + Q(t) dt$, and a function $f(t, x)$ that is $C^{1,2}$ (once differentiable in t , twice in x):

$$df(t, X(t)) = f_t(t, X(t)) dt + f_x(t, X(t))P(t) dB(t) + f_x(t, X(t))Q(t) dt + \frac{1}{2}f_{xx}(t, X(t))P^2(t) dt.$$

Proof: The Taylor expansion of $f(t, x)$ in two variables gives:

$$df(t, X(t)) = f_t dt + f_x dX(t) + \frac{1}{2}f_{xx} (dX(t))^2.$$

Using $(dX)^2 = P^2 dt$ and $dX = Q dt + P dB$ yields the stated formula directly. $\square$

This can be written more compactly as:

$$df = \left(f_t + Qf_x + \frac{1}{2}P^2f_{xx} \right) dt + Pf_x dB.$$

5.5 Applications of Itô's Formula

Example – Stochastic Differentials via Itô's Formula

Compute $dY(t)$ for each of the following:

(a) $Y(t) = e^{\beta t}$ (no Brownian motion component, so $P = 0$, $X = t$):

$$dY = \beta e^{\beta t} dt = \beta Y(t) dt.$$

(b) $Y(t) = e^{\beta B(t)}$ (apply Itô with $f(x) = e^{\beta x}$, $dX = dB$, so $P = 1$, $Q = 0$):

$$dY = \beta e^{\beta B} dB + \frac{1}{2} \beta^2 e^{\beta B} dt = \beta Y dB + \frac{\beta^2}{2} Y dt.$$

(c) $Y(t) = e^{\beta X(t)}$ where $dX = \mu dt + \sigma dB$:

$$dY = \beta Y dX + \frac{1}{2} \beta^2 Y (dX)^2 = \left(\beta \mu + \frac{\beta^2 \sigma^2}{2} \right) Y dt + \beta \sigma Y dB.$$

(d) $Y(t) = 1/X(t)$ where $dX = \alpha X dt + \sigma X dB$ (GBM). Here $f(x) = 1/x$, $f' = -1/x^2$, $f'' = 2/x^3$:

$$dY = -\frac{1}{X^2} dX + \frac{1}{2} \cdot \frac{2}{X^3} (dX)^2 = (-\alpha + \sigma^2) Y dt - \sigma Y dB.$$

Example – Verification of an SDE

Show that $Y(t) := t^2 B(t)^3$ satisfies:

$$dY(t) = \left(\frac{2Y(t)}{t} + 3(t^4 Y(t))^{1/3} \right) dt + 3(tY(t))^{2/3} dB(t), \quad Y(0) = 0.$$

Solution: Set $f(t, x) = t^2 x^3$. Then:

$$f_t = 2tx^3, \quad f_x = 3t^2 x^2, \quad f_{xx} = 6t^2 x.$$

With $dX = dB(t)$ (so $P = 1$, $Q = 0$), Itô's formula gives:

$$\begin{aligned} dY &= f_t dt + f_x dB + \frac{1}{2} f_{xx} dt \\ &= 2tB^3 dt + 3t^2 B^2 dB + 3t^2 B dt. \end{aligned}$$

Rewriting in terms of $Y = t^2 B^3$: note $B^3 = Y/t^2$, $B^2 = (Y/t^2)^{2/3} \cdot t^{2/3} = (t^4 Y)^{1/3}/t^2 \cdot t = (t^4 Y)^{1/3}/t$, and $B = (tY)^{1/3}/t^{2/3}$. Substituting confirms the result. $\square$

6 Geometric Brownian Motion

6.1 The Bachelier Model

Before the modern (GBM) approach, **Louis Bachelier** modeled stock prices as:

$$dS(t) = \sigma dB(t).$$

Properties of the Bachelier model:

- The solution is $S_t = S_0 + \sigma B_t$.
- Price *increments*: $S_t - S_{t-1} = \sigma(B_t - B_{t-1}) \sim \mathcal{N}(0, \sigma^2)$.
- Increments are **independent**: $S_t - S_{t-1}$ for $t = 1, 2, \dots$ are i.i.d. normal.
- The volatility estimate: $\hat{\sigma} = \text{sd}(S_t - S_{t-1})$.

Remark

A major drawback of the Bachelier model is that it allows stock prices to become **negative**, since normal random variables take negative values. The Geometric Brownian Motion (GBM) model resolves this.

6.2 The GBM Stochastic Differential Equation

The **Geometric Brownian Motion** (Black–Scholes model) governs stock prices via:

$$dS(t) = \mu S(t) dt + \sigma S(t) dB(t),$$

where:

- μ is the **drift** (expected rate of return),
- $\sigma > 0$ is the **volatility**,
- $S(t) > 0$ for all t (prices remain positive).

6.3 Solving the GBM SDE via Itô's Formula

To solve the GBM SDE, we use the **log-transformation**. Choose $f(x, t) = \ln x$. Then $f_t = 0$, $f_x = 1/x$, $f_{xx} = -1/x^2$.

Identifying $P(t) = \sigma S(t)$ and $Q(t) = \mu S(t)$, Itô's formula gives:

$$\begin{aligned} d \ln S(t) &= \frac{1}{S(t)} \cdot \sigma S(t) dB + \frac{1}{S(t)} \cdot \mu S(t) dt + \frac{1}{2} \cdot \frac{-1}{S(t)^2} \cdot \sigma^2 S(t)^2 dt \\ &= \sigma dB(t) + \left(\mu - \frac{\sigma^2}{2} \right) dt. \end{aligned}$$

Integrating both sides from 0 to t :

$$\ln S(t) - \ln S(0) = \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma B(t).$$

Key Formula: GBM Solution

$$S(t) = S(0) \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma B(t) \right].$$

This ensures $S(t) > 0$ for all $t \geq 0$.

6.4 Log>Returns and Distribution

Define the **log-return** over one period as:

$$r(t) = \log \frac{S(t)}{S(t-1)} = \left(\mu - \frac{\sigma^2}{2} \right) + \sigma (B(t) - B(t-1)).$$

Statistical Properties of Log>Returns

Under GBM, log-returns $r(t)$ satisfy:

- $r(t) \sim \mathcal{N}\left(\mu - \frac{\sigma^2}{2}, \sigma^2\right)$ for each t .
- $r(1), r(2), \dots, r(T)$ are **independent** (from independent Brownian increments).

6.5 Parameter Estimation

Given a time series of prices $S(0), S(1), \dots, S(T)$, we estimate the GBM parameters as follows.

Volatility: The variance of log-returns equals σ^2 , so:

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (r(t) - \bar{r})^2 \quad (\text{sample variance of log-returns}).$$

Drift: The mean log-return equals $\mu - \sigma^2/2$, so:

$$\widehat{\mu - \frac{\sigma^2}{2}} = \bar{r}(t) \implies \hat{\mu} = \bar{r}(t) + \frac{\hat{\sigma}^2}{2}.$$

Example – Parameter Estimation in Practice

Step 1: Compute daily log-returns: $r(t) = \ln S(t) - \ln S(t-1)$.

Step 2: Estimate $\hat{\sigma}^2$ = sample variance of $r(t)$.

Step 3: Estimate $\hat{\mu} = \bar{r} + \hat{\sigma}^2/2$.

Under GBM, **all parameters can be estimated from daily log-returns** of the equity.

6.6 Simulation of GBM

Given $S(t) = S(0) \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B(t)\right]$, we simulate as follows:

Algorithm: Simulating GBM

Step 1. Discretize $[0, T]$ into $N = T/\Delta t$ uniform sub-intervals of length Δt .

Step 2. Generate N i.i.d. normal random numbers with mean 0 and variance $\sigma^2 \Delta t$.

These simulate the Brownian increments $\sigma(B((j+1)\Delta t) - B(j\Delta t))$.

Step 3. Take the **cumulative sum** of these random numbers. Call this vector **p1**.

This approximates $\sigma B(j\Delta t)$ for $j = 1, \dots, N$.

Step 4. Compute $p_{2,j} = \left(\mu - \frac{\sigma^2}{2}\right)j\Delta t$ for $j = 1, \dots, N$.

Call this vector **p2**.

Step 5. The GBM path is: $S(j\Delta t) = S(0) e^{p_{1,j} + p_{2,j}}$.

7 Solving Stochastic Differential Equations

7.1 The Ornstein–Uhlenbeck Process

The **Ornstein–Uhlenbeck (OU)** process is one of the most important SDEs, widely used in physics (mean-reverting noise) and finance (interest rate models).

Definition – OU Process

The Ornstein–Uhlenbeck process satisfies the SDE:

$$dX(t) = -\lambda X(t) dt + \sigma dB(t),$$

where $\lambda > 0$ is the **mean-reversion rate** and $\sigma > 0$ is the **volatility**.

7.2 Solving the OU SDE

We use the **integrating factor** technique, motivated by Itô's formula. Consider the substitution $Y(t) = e^{\lambda t} X(t)$, i.e., $f(x, t) = e^{\lambda t} x$.

With this, $f_t = \lambda e^{\lambda t} x$, $f_x = e^{\lambda t}$, $f_{xx} = 0$. Applying Itô's formula (with $P = \sigma$, $Q = -\lambda X$):

$$\begin{aligned} dY(t) &= \lambda e^{\lambda t} X(t) dt + e^{\lambda t} (-\lambda X(t)) dt + e^{\lambda t} \sigma dB(t) \\ &= e^{\lambda t} \sigma dB(t). \end{aligned}$$

The drift terms cancel! Integrating:

$$Y(t) = Y(0) + \sigma \int_0^t e^{\lambda s} dB(s).$$

Since $Y(t) = e^{\lambda t} X(t)$, we recover:

Key Formula: OU Process Solution

$$X(t) = X(0) e^{-\lambda t} + \sigma e^{-\lambda t} \int_0^t e^{\lambda s} dB(s).$$

7.3 Mean and Variance of the OU Process

Using the zero-mean property of the Itô integral:

Mean:

$$\mathbb{E}[X(t)] = X(0) e^{-\lambda t}.$$

As $t \rightarrow \infty$, $\mathbb{E}[X(t)] \rightarrow 0$ — the process reverts to zero (**mean reversion**).

Variance: Using the Itô isometry:

$$\begin{aligned}\text{Var}(X(t)) &= \mathbb{E} \left[\left(\sigma e^{-\lambda t} \int_0^t e^{\lambda s} dB(s) \right)^2 \right] = \sigma^2 e^{-2\lambda t} \int_0^t e^{2\lambda s} ds \\ &= \sigma^2 e^{-2\lambda t} \cdot \frac{e^{2\lambda t} - 1}{2\lambda} = \frac{\sigma^2}{2\lambda} (1 - e^{-2\lambda t}).\end{aligned}$$

Long-Run Behavior of the OU Process

As $t \rightarrow \infty$:

$$\mathbb{E}[X(t)] \rightarrow 0, \quad \text{Var}(X(t)) \rightarrow \frac{\sigma^2}{2\lambda}.$$

The OU process converges in distribution to $\mathcal{N}\left(0, \frac{\sigma^2}{2\lambda}\right)$ — its **stationary distribution**.

8 Summary of Key Results

Concept	Key Formula / Result
Itô Integral (mean)	$\mathbb{E} \left[\int_0^t X dB \right] = 0$
Itô Isometry	$\mathbb{E} \left[\left(\int_0^t X dB \right)^2 \right] = \int_0^t \mathbb{E}[X^2] ds$
Itô's Formula	$df(X) = f'(X) dX + \frac{1}{2} f''(X) P^2 dt$
Extended Itô	$df(t, X) = f_t dt + f_x dX + \frac{1}{2} f_{xx} P^2 dt$
GBM Solution	$S(t) = S(0) \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma B(t) \right]$
OU Solution	$X(t) = X(0) e^{-\lambda t} + \sigma e^{-\lambda t} \int_0^t e^{\lambda s} dB(s)$
OU Variance	$\text{Var}(X(t)) = \frac{\sigma^2}{2\lambda} (1 - e^{-2\lambda t})$

9 References

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